Digital Image Processing (DIP) Course Project Guidelines

Project Team: 1–2 Students | Topic Registration: Online | Deadline: Sunday, 18 October 2026

IMPORTANT: Students may select a topic from the supplied 62-project Industry-Linked Advanced AI Project Bank, adapt one of those ideas to DIP, or propose an entirely original topic. The proposed project must have a clear and substantive connection to Digital Image Processing and/or image/vision AI taught in the course. If a team has two students, BOTH students must submit the online topic form separately, mentioning the same project title and partner details.

1. Topic Selection and Registration

- A project may be done individually or by a team of two students.
- Students are free to propose their own project topic. It need not come from the 62-project list.
- A self-proposed topic must show a genuine DIP/vision connection: image acquisition/pre-processing, enhancement, filtering, edge/feature extraction, segmentation, morphology, colour processing, object/region analysis, or a related AI/CV component such as CNN, object detection, Vision Transformer (ViT), multimodal/VLM, etc.
- The 62-project bank is an inspiration/reference list. The named companies are industry references only; the topics are not claimed to be officially assigned or sponsored by those companies.
- Topic submission deadline: Sunday, 18 October 2026, through the announced online form.
- For a two-student group, each student must submit the form separately. Both submissions must carry identical project title/ID and clearly mention the partner's name and roll number.
- Final topic approval may require refinement if the proposed work has insufficient DIP content or substantially duplicates another team's topic.

2. What Counts as a DIP Course Project?

The project should contain an end-to-end image/vision pipeline, not only a generic app, dashboard, LLM chat interface, or a pre-trained model called without analysis. A suitable project normally contains:

- Image/video/remote-sensing/document-image input or a clearly defined visual dataset.
- A meaningful DIP stage: enhancement, histogram/contrast operation, denoising/filtering, interpolation, colour-space processing, edge/feature extraction, thresholding/segmentation, morphology, connected components/region features, change detection, etc.
- Where appropriate, a taught-related AI/CV stage such as classical ML, CNN, object detection/segmentation, ViT/Transformer, image embeddings, VLM/multimodal model, or generative vision.
- A baseline and at least one improved/alternative method, with visual and quantitative comparison.
- A clear output: detected/segmented/classified/restored/enhanced image, quality score, map, bounding boxes, labels, change mask, retrieval result, or another measurable visual result.
- Reproducible code and documented data. Public/open data, student-captured data with appropriate consent, or carefully generated synthetic data may be used.

3. Minimum Technical Expectations

Stage	Minimum Expectation	Evidence in Final Work
Problem & Data	Define the visual problem, input type and dataset/source.	Sample inputs, dataset description and problem statement.
DIP Core	Use one or more DIP operations meaningfully; do not add them only for formality.	Intermediate images/results showing what each operation contributes.
AI/CV Link	Where relevant, connect classical DIP to a learnt vision method.	Model/architecture, features/embeddings, training/inference or comparison.

Comparison	Compare baseline vs improved/alternative pipeline.	Metrics + visual examples; discuss success and failure cases.
Implementation	Working MATLAB/Python/other suitable implementation.	Executable code/notebook/application and reproducible instructions.
Outcome	Generate an interpretable visual/quantitative output.	Screenshots/figures/tables plus concise technical analysis.

4. Online Topic Submission – What Each Student Must Submit

- Student name and roll number.
- Team type: Individual / Two-student team.
- Partner name and roll number (if applicable).
- Project title.
- Source: one of the 62 listed projects (mention Project ID) OR “Own Topic”.
- 3–5 line problem statement.
- 3–5 specific DIP/course connections that will actually be implemented.
- Expected input/data and expected output.
- Planned comparison/evaluation metric(s) or measurable result.

5. DIP Connections Already Present in the 62-Project Bank

The following entries already contain an explicit image/vision/DIP-related hook in the supplied project statement. Students may take inspiration from these and re-scope them as a DIP course project. Projects not listed below are primarily data/NLP/finance/IoT/software/agentic projects as presently written; they may still be proposed only if the student adds a clear, technically justified image/vision component.

ID	Project (short)	DIP Link	Where the link appears in the supplied bank	Suitable DIP emphasis for this course
004	Insurance Document Intake / Claim Readiness	Direct	Scanned forms, invoices and IDs; OCR + document classification.	Document enhancement, grayscale/contrast, thresholding, morphology, region/character segmentation, OCR accuracy comparison.
013	3D Commerce Scene Planning & Asset Composition	Vision extension	Product images + vision-language tagging + 3D assets.	Image preprocessing/segmentation, feature or ViT/VLM embeddings, product-image retrieval/recognition.
014	WebAR Asset Optimization & Adaptive Delivery	Related DIP	Textures and visual-quality optimisation for 3D/WebAR assets.	Image resizing/interpolation, texture quality, resolution/compression-quality trade-off, visual quality metrics.
016	Vision-Based Exercise Form Analysis	Direct	Recorded exercise videos; pose estimation + temporal classification.	Frame preprocessing, foreground/pose extraction, features/keypoints, CNN/pose model, temporal analysis.
029	Multimodal eKYC Verification	Direct	Face/liveness/document data; face matching + OCR + liveness/deepfake checks.	Face/document enhancement, segmentation, feature extraction, CNN/ViT, image-quality and verification metrics.
033	Multi-Sensor Disaster Change Detection	Direct	Sentinel-1 SAR + Sentinel-2 optical imagery; segmentation/change detection.	Multiband/FCC processing, registration/preprocessing, threshold/change maps, segmentation, CNN/ViT fusion.
034	Air-Gapped Geospatial Inference	Direct	Large raster tiles + trained CV models; tiling/batching/inference.	Raster tiling, normalization, patch processing, segmentation/detection, stitching and accuracy/speed analysis.
042	Application Readiness / Document Verification	Direct	Transcripts/application documents; OCR/NER.	Document image preprocessing, contrast/thresholding, morphology, OCR and field extraction.
046	Multimodal CPG Concept Testing	Vision extension	Product images + packaging text; vision-language embeddings.	Colour/texture/shape preprocessing, image features, CNN/ViT/VLM embeddings, visual similarity/retrieval.
049	Camera-Only Road Scene Understanding	Direct	Driving imagery; detection, segmentation and lane perception.	Edges/Hough vs learnt lane detection, object detection, semantic segmentation, CNN/ViT comparison.
050	Autonomous-Driving Domain-Shift Adaptation	Vision extension	Driving datasets split by city/weather/time/domain.	Brightness/contrast/weather normalization, augmentation, image-quality/domain analysis, CNN/ViT robustness.
055	Fashion Creative Generation / Revision	Vision extension	Garment images, poses, captions; image generation/inpainting.	Segmentation/masks, resizing/colour consistency, inpainting, perceptual/similarity evaluation.
056	E-Commerce Image Quality Inspection	Direct	Product images with synthetic defects; image-quality + background/object/layout compliance.	Contrast/sharpness/noise measures, segmentation, morphology, object/background analysis, quality classification.

057	Creative Performance Prediction / Variant Selection	Vision extension	Ad creatives; multimodal embeddings + performance prediction.	Image feature/embedding extraction, colour/layout/quality analysis, CNN/ViT/VLM features linked to prediction.

6. Recommended DIP Project Design Pattern

A strong project should make the contribution of DIP visible. One recommended structure is:

Input Image/Video → Pre-processing/Enhancement → Segmentation/Features → AI/CV Model (if used) → Visual Output → Quantitative Evaluation → Failure Analysis

- Do not hide the entire project behind a single pre-trained API call. Show the input, intermediate DIP stages, final result and why the processing choices matter.
- Prefer a comparison such as classical DIP vs learned method, method A vs method B, before/after preprocessing, different thresholds/filters/SEs, CNN vs ViT features, or baseline vs proposed improvement.
- Use appropriate metrics: accuracy/F1/IoU/Dice for classification or segmentation; precision/recall for detection; PSNR/SSIM or task-specific quality measures for restoration/enhancement; runtime/memory when relevant.

7. Minimum Final Deliverables

- Working source code and a reproducible execution procedure (Git repository preferred).
- Dataset/source description and preprocessing details.
- Block diagram / architecture showing where DIP operations and AI/CV components are used.
- Baseline and improved/alternative method with quantitative comparison.
- Representative screenshots/figures of input, intermediate DIP stages and final output.
- Analysis of at least a few success and failure cases—not only screenshots.
- Final report and short demonstration/presentation. For a two-student team, both members must be able to explain the complete pipeline and clearly state their individual contribution.

8. Important Note on the 62-Project Industry References

The supplied Advanced AI Project Bank states that the company name is an industry reference indicating a related problem domain; it does not mean that the project is officially assigned, sponsored or approved by that company. Students should use public/open data or a documented synthetic dataset unless approved data access is separately available. For this DIP course, the 62 topics are inspiration—not a requirement to reproduce the full Agentic AI architecture. The mandatory requirement is a defensible, substantive DIP/image/vision contribution.

Reference basis: "Advanced AI Course – Industry-Linked Agentic AI Project Bank" (62 supplied project statements).